Phillip Lippe

PhD student focusing on (causal) representation learning and generative pretraining with experience in large-scale training. Looking for Research Scientist positions based in Europe and US starting from October 2024.

Nationality **German** Languages **German (native), English (C1)** **Amsterdam, The Netherlands**

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EXPERIENCE

Student Researcher

Google DeepMind

Aug 2023 – Nov 2023 Amsterdam, Netherlands

LLM Pretraining | Diffusion Models | Scaling

- Research in generative multimodal pretraining for image-text sequences; hosted by Mostafa Dehghani.
- Training up to 4B parameter models over 512 devices.
- Results internally used in the Gemini foundation model.

Research Intern

Microsoft Research

Mar 2023 – May 2023 Amsterdam, Netherlands

Diffusion Models | Video Generation | AI4Science

- Research in neural PDE solvers for scientific simulation; hosted by Johannes Brandstetter.
- Published *PDE-Refiner: Achieving Accurate Long Rollouts with Neural PDE Solvers* at NeurIPS 2023 (pdf).

Student Research Intern

Daimler AG, Mercedes-Benz (MBRDNA)

Dec 2016 – Sep 2018 Sunnyvale, United States
Stuttgart, Germany

Computer Vision | Generative Models | Scrum

- Three research internships on computer vision and deep generative models for autonomous driving.
- Performed agile software development with Scrum.

PROJECTS

Google Developer Expert, Machine Learning

- Member of the ML GDE program for expertise and public content sharing in JAX+Flax since August 2022.
- Presented tutorials on distributed training for scaling deep learning models to billion parameters.

Tutorials

- Created and taught various Deep Learning tutorial notebooks in PyTorch and JAX (website, ∼40k page visits per month, >2k GitHub stars). Part of the UvA DL course and official tutorials of PyTorch Lightning.

Lecturer and Teaching Assistant

- Given lectures in the graduate courses Deep Learning, Foundation Models, and Advanced NLP at the University of Amsterdam (2019-2024).

EDUCATION

PhD, Artificial Intelligence

University of Amsterdam, QUVA lab

Sep 2020 – present Amsterdam, Netherlands

- PhD Advisors: dr. Efstratios Gavves and dr. Taco Cohen.
- My research focuses on causality and ML, such as causal representation learning from temporal data.
- ELLIS PhD; collaboration with Qualcomm AI Research.
- Expected graduation date: September 2024.

Master of Science, Artificial Intelligence

University of Amsterdam

Sep 2018 – Aug 2020 Amsterdam, Netherlands

- Courses on AI, including ML, DL, NLP, IR, CV, and RL.
- Thesis on "Categorical Normalizing Flows" (publication), applied for graph/molecule generation.
- Final GPA: 9.5, cum laude (Dutch grading system).

Bachelor of Engineering, Computer Science

Baden-Wuerttemberg Cooperative State University

Oct 2015 – Sep 2018 Stuttgart, Germany

- Cooperative study program in cooperation with Daimler AG/Mercedes-Benz R&D for autonomous driving.
- Final GPA: 1.0 (German grading system).

AWARDS

- Google-internal Peer Award for finding and resolving severe bug in distributed data loading, affecting hundreds of large-scale research experiments globally.
- Won 4th place in the NeurIPS 2020 challenge "Hateful Memes" of Facebook AI (paper).
- Best CS undergraduate student at the Baden-Wuerttemberg Cooperative State University 2018.
- Awarded SchülerUni Scholarship 2012/13 for taking university courses during high school.

SKILLS

DL Frameworks: PyTorch (2018-present, R&T), JAX/Flax (2021-present, R&T), Tensorflow (2017-2023, R), Caffe (2016-2017, R) (R - Research, T - Teaching)

Programming languages: Python (2016-present), HTML (2012-present), Java (2012-2019), C (2015-2017)

Additional skills: SLURM cluster computing (2018-present), git (2015-present), Docker (2017-present)

SELECTED PUBLICATIONS (FULL LIST OF PUBLICATIONS)

- Samuele Papa, Riccardo Valperga, David M Knigge, Miltiadis Kofinas, Phillip Lippe, Jan-jakob Sonke, Efstratios Gavves: **How to Train Neural Field Representations: A Comprehensive Study and Benchmark**. The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024 (link)
  — *Own contributions:* Core contributor of library development in JAX, proposed and optimized vmap parallelization, advised in model development.
- Davide Talon, Phillip Lippe, Stuart James, Alessio Del Bue, Sara Magliacane: **Towards the Reusability and Compositionality of Causal Representations**. Third Conference on Causal Learning and Reasoning (CLeaR), 2024 (link) **[Oral]**
  — *Own contributions:* Worked on model development, experiment design, and theoretical results.. Contributed to writing.
- Phillip Lippe, Bastiaan S. Veeling, Paris Perdikaris, Richard E. Turner, Johannes Brandstetter: **PDE-Refiner: Achieving Accurate Long Rollouts with Neural PDE Solvers**. Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023 (link). **[Spotlight]**
- Sindy Löwe, Phillip Lippe, Francesco Locatello, Max Welling: **Rotating Features for Object Discovery**. Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023 (link). **[Oral]**
  — *Own contributions:* Advised in model development, experiment design and model optimization. Contributed to writing.
- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **BISCUIT: Causal Representation Learning from Binary Interactions**. The 39th Conference on Uncertainty in Artificial Intelligence (UAI), 2023 (link). **[Spotlight]**
- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **Causal Representation Learning for Instantaneous and Temporal Effects in Interactive Systems**. International Conference on Learning Representations (ICLR), 2023 (link).
- Adeel Pervez, Phillip Lippe, Efstratios Gavves: **Differentiable Mathematical Programming for Object-Centric Representation Learning**. International Conference on Learning Representations (ICLR), 2023 (link).
  — *Own contributions:* Proposed application to object-centric learning. Advised in model development and experimental setups. Contributed to writing.
- Johann Brehmer, Pim de Haan, Phillip Lippe, Taco Cohen: **Weakly supervised causal representation learning**. Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), 2022 (link).
  — *Own contributions:* Advised in theory and model development, created datasets, implemented causal discovery pipeline, and contributed to writing.
- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **CITRIS: Causal Identifiability from Temporal Intervened Sequences**. International Conference on Machine Learning (ICML), 2022 (link). **[Spotlight]**
  — *Own contributions:* Advised in training and finetuning of large language Transformers, guided as teaching assistant through the project.
- Phillip Lippe, Taco Cohen and Efstratios Gavves: **Efficient Neural Causal Discovery without Acyclicity Constraints**. International Conference on Learning Representations (ICLR), 2022 (link).
- Phillip Lippe and Efstratios Gavves: **Categorical Normalizing Flows via Continuous Transformations**. International Conference on Learning Representations (ICLR), 2021 (link).

I have served as reviewer for the following venues: ECCV-2020, ICCV-2021, CausalUAI-2021, ICLR-2022, CVPR-2022, CLeaR-2022, NeurIPS-2022, CRL-2022, CML4Impact-2022, CDS-2022, nCSI-2022, ICLR-2023, CLeaR-2023, TSRL4H-2023, Physics4ML-2023, UAI-2023, NeurIPS-2023, ICML-2024. I co-organize the CORR workshop at CVPR-2024.

SELECTED PUBLIC TALKS (FULL LIST OF TALKS)

03/2024 Invited Talk at the Deep Thinking Hour (UvA) on "Training Models at Scale"
02/2024 Invited Talk at the CARE Talk Series by Valencelabs on "BISCUIT: CRL from Binary Interactions"
02/2024 Invited Talk at the Bellairs Workshop on Causality on "On Practical Challenges of Scaling CRL"
09/2023 Invited Talk at the AI4Science Talk Series on "Achieving Accurate Long Rollouts with Neural PDE Solvers"
02/2023 Invited Talk at the Rising Stars in AI Symposium at KAUST on "Causal Representation Learning"
12/2022 Invited Talk at the Google Student Developer Club, University of Augsburg on "ML with JAX and Flax"
08/2022 Invited Talk at the First Workshop on Causal Representation Learning at UAI 2022 on "Learning Causal Variables from Temporal Sequences with Interventions"

SELECTED OPEN-SOURCE CONTRIBUTIONS (VERSION: APRIL 12, 2024)

phlippe/uvadlc_notebooks — 2,114 stars, 512 forks

This repository contains a series of Deep Learning tutorials written in PyTorch and JAX (Jupyter notebooks). The topics range from introductions to PyTorch/JAX, Optimization and Initialization, to recent advances in (Vision) Transformers, Generative Models, and Distributed Training at Scale. The tutorials are easily accessible and executable via its website. So far, 40 tutorials have been authored by myself, and 18 notebooks by collaborators.

Lightning-AI/tutorials — 267 stars, 71 forks

This repository contains the official tutorials for PyTorch Lightning. In collaboration with the Lightning-AI team, I have integrated my tutorials from the phlippe/uvadlc_notebooks repository. This included common steps for large, open-source libraries, such as code reviews, PEP8 formatting, automatic github actions, and similar.

phlippe/jax_trainer — 16 stars, 1 forks

This repository contains a Lightning-like Trainer API for JAX with Flax. While mainly used in my own projects, it contains common practices for reproducible experimentation in deep learning and optimized for single-device environments.